BCMID — Breast Cancer Multimodal Intelligent Diagnosis

Deep Learning technical report • English version • ESPRIT academic project • 2026–2027

Executive summary and scope

BCMID is a research prototype for supporting the interpretation of breast imaging studies by combining mammography, ultrasound and clinical notes. It connects a ResNet-18 vision component, Grad-CAM explanations, retrieval-augmented generation (RAG), LLM-based synthesis and a deterministic safety review before report production. Its intended role is clinical decision support under human supervision, not autonomous diagnosis.

Technical objectives

(1) Validate and trace input data; (2) assess the contribution of each imaging modality and their fusion; (3) produce interpretable, evidence-linked descriptions; (4) constrain unsupported generative output; and (5) generate structured reports requiring clinician review.

Evidence and reporting approach

This revision retains the original report’s design while checking the supplied Python source, tests, FAISS indices and project artifacts. Reported experiment results are distinguished from implemented functionality and from runtime behavior that cannot be independently established. Probability-to-BI-RADS conversion in the repository is an unvalidated heuristic.

Report structure

1. Problem and architecture • 2. Tool choices • 3. LLM choice • 4. RAG design • 5. Evaluation • 6. Security and limitations • Conclusion.

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Architecture overview

The full-size architecture in Appendix A reflects the user-supplied diagram: input modalities, data-quality checks, multimodal imaging analysis, clinical RAG, reasoning, safety validation and final reporting, orchestrated through n8n. The diagram also contains illustrative technology options; the report sections distinguish these from specific components identified in the codebase.

How to read this report

Numbers from the earlier report are distinguished from results independently reproducible from the delivered artifacts. BCMID remains an academic research prototype that has not been validated for clinical use.

1. Agent architecture and rationale

The system is organized into seven specialized processing blocks to separate predictive modeling, medical document retrieval, natural-language synthesis and deterministic checks. This decomposition supports modular development, error isolation and traceable review, which are particularly important for heterogeneous clinical inputs.

Block	Role	Primary output
1. Ingestion	Validate and identify input files	Ingestible study
2. Data Quality	De-identification, integrity, preprocessing	Clean data / manifest
3. Vision	ResNet-18 scoring and explanation	Scores, uncertainty, Grad-CAM
4. Clinical RAG	Retrieve guidelines and precedents	Attributed evidence excerpts
5. Reasoning	Synthesize vision results and evidence	Draft clinical narrative
6. Safety	Check claims, numbers and references	Approval or revision feedback
7. Reporting	Render human-readable artifacts	PDF / HTML report

End-to-end flow

Mammography + ultrasound + clinical notes → ingestion and quality control → vision and RAG → LLM-assisted reasoning → safety audit → report for clinician sign-off. A supplied n8n workflow supports event routing. Technically, this is an orchestrated multi-component pipeline; autonomous decision-making should not be attributed to every stage.

Why this architecture fits

Medical images and textual guidelines require different representations. Separating perception, knowledge retrieval, synthesis and verification enables targeted testing and avoids placing all responsibility on an opaque language model. The Safety-to-Reasoning feedback loop allows up to two revisions; unresolved inconsistencies should trigger manual review.

2. Tools and technical choices

Layer	Selected technologies	Rationale and trade-off
Vision	PyTorch, torchvision, ResNet-18	Relatively compact CNN; Grad-CAM support; requires verifiable trained weights.
Image quality	OpenCV, Pillow, OCR where configured	Cleaning and artifact reduction; must preserve clinical findings.
Retrieval	FAISS, sentence-transformers	Efficient local similarity search; source and index governance remain necessary.
Services	FastAPI, Pydantic	Typed API and validation; network protection remains a deployment requirement.
Generation	Transformers, Qwen2.5-3B-Instruct	Potential local inference; documented fallback when unavailable.
Outputs	ReportLab, HTML	Structured printable and web reports.
Automation	n8n	Webhook-based workflow routing; not evidence of clinical integration certification.
Testing	pytest	Unit and integration checks for quality and safety logic.

Selection trade-offs

ResNet-18 prioritizes moderate compute cost over very large visual backbones, but requires domain-specific validation and calibration. FAISS supports on-premises deployments, while introducing local index management. n8n improves workflow visibility but expands the attack surface. Local processing can reduce third-party data transfer; it does not, by itself, guarantee legal compliance.

3. Language model selection

Qwen2.5-3B-Instruct and its responsibility

The implementation in src/reasoning/reasoning_agent.py targets Qwen/Qwen2.5-3B-Instruct. Its purpose is to turn structured imaging outputs and retrieved evidence into a clinical draft. Image classification remains the vision system’s responsibility, and final interpretation belongs to a qualified professional.

The model was chosen for its approximately 3-billion-parameter scale, plausible local deployment, instruction-following capability and structured text generation. Quantization could lower memory use, but should be benchmarked for quality degradation. The code uses decoding with do_sample=False, which reduces output variability without ensuring factual correctness.

Real inference versus rule-based fallback

The module attempts local loading through Transformers (local_files_only=True). When weights are missing or loading fails, it uses mock_reasoning_agent_llm, tagging responses with reasoning_mode=rule_based_fallback and simulation_mode=True. Such results must not be represented as measured Qwen inference.

Constrained synthesis

Prompts combine case findings and retrieved passages, while safety checks inspect the generated claims, numerical consistency and references. The architecture is designed to request a bounded revision or a clinician review when outputs cannot be grounded. More controlled tests are needed to quantify hallucination reduction.

4. Clinical retrieval-augmented generation

RAG instead of knowledge-only prompting

RAG enables evidence retrieval from a versioned clinical corpus without retraining the LLM after each knowledge update. It allows the generated explanation to point to supporting passages. Retrieval can reduce unsupported statements, but does not guarantee medically correct answers.

Corpus, embeddings and retrieval

The archive includes acr_guidelines_index.faiss and rag_index.faiss with accompanying data. The earlier report describes 11 guideline rules and 317 historical case fragments. The selected encoder is BAAI/bge-small-en-v1.5 (384-dimensional embeddings). A case query is encoded, matched with relevant passages and a small top-k evidence set (typically four) is passed to the reasoning component.

The ikbelwork/clinical_rag_enhancer.py module supports patient-ID exclusion (exclude_pseudo_id) during retrieval. This feature should not be conflated with proof that all cross-fold leakage is prevented. The implementation includes alternative retrieval behavior if embeddings are unavailable. Guidelines and historical case documents should be distinguished in prompts and output citations.

RAG-specific evaluation

Build clinician-annotated retrieval queries; measure Recall@4, citation precision, relevance of retrieved segments and unsupported statement rate. Record document versions, permissions and source provenance. An included index should not automatically be described as an officially licensed guideline database.

5. System evaluation

The original report provides the following results on a stated cohort of 323 paired patients. These are previously reported numbers: the delivered ZIP does not include every trained checkpoint, prediction artifact and execution setting needed for a fully independent reproduction.

Configuration	Reported ROC AUC	Reported sensitivity
ResNet-18 mammography	0.7917	76.2%
ResNet-18 ultrasound	0.6743	68.4%
Multimodal fusion	0.8198	83.1%

Explainability and software testing

The earlier report describes 387 Grad-CAM maps and 87.4% of activation energy inside breast tissue. Spatial concentration is useful diagnostic information, but does not prove that a classification is medically correct. The tests/ directory contains checks for anonymization, file integrity, modality completeness, splits, safety and inference logic. The earlier document records 61 passing tests; this revision does not claim a fresh independent run.

Critical interpretation of findings

No ResNet-18 *.pth/*.pt checkpoint was found in the supplied archive. src/vision/vision_agent.py explicitly falls back to simulated, metadata-based scores if no checkpoint is loaded. Reported model benchmarks therefore need to be separated from local simulation demonstrations. Before clinical use, conduct external validation, confidence-interval analysis, sensitivity at clinically meaningful thresholds, subgroup analyses and error audits.

6. Safety, privacy and regulatory considerations

Data protection and input integrity

The project documents pseudonymous identifiers, clinical text cleaning, SHA-256 fingerprints and patient-aware splitting. Hashed pseudonyms may still be linkable to individuals; appropriate organizational and technical controls include data minimization, strong authentication, authorization, encrypted storage and transit, logging, secret management and retention rules.

Output safety and mandatory human oversight

The Safety Agent checks source reference validity, lexical claims, agreement with vision outputs and uncertainty statements. A failed check should result in clinician review instead of definitive automated advice. The repository’s thresholds mapping malignancy scores to BI-RADS are heuristic and must not be treated as clinically validated equivalents.

Threat model and hardening roadmap

Test prompt injection in clinical notes and RAG documents, malformed image inputs, training/retrieval leakage, corpus poisoning, report authorization bypass, exposed n8n webhooks and model drift. Enforce role-based access, input validation, provenance checks, audit trails, component versioning and incident procedures.

Regulatory interpretation

BCMID should be described as an academic research prototype, not validated for clinical use. Whether the system falls within EU MDR 2017/745, and any applicable risk class, depends on a formal analysis of its intended purpose and jurisdiction. Software safety controls alone do not amount to GDPR, HIPAA or MDR compliance.

Conclusion

BCMID demonstrates a well-structured approach to connecting multimodal learning, evidence retrieval, generated interpretation and software-based safeguards. Priorities are to supply trained vision checkpoints, reproduce reported performance, assess real Qwen inference, audit RAG quality and obtain expert clinical review before any clinical deployment claim.

Primary sources: user-provided BCMID v3.0 report and GitHub project archive (README.md, src/vision/vision_agent.py, src/reasoning/reasoning_agent.py, ikbelwork/clinical_rag_enhancer.py, ikbelwork/run_orchestrator.py, tests/, bcmid_work/).

APPENDIX A — Overall BCMID project architecture

